Journal of the Optical Society of America B

OPTICAL PHYSICS

On-chip supercontinuum generation: from recent advances to applications [Invited]

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Received 14 August 2025; revised 9 October 2025; accepted 13 October 2025; posted 15 October 2025; published 10 December 2025

Nonlinear frequency conversion is crucial for reaching new optical frequencies and creating light sources that are essential for research and industrial applications. Among the various strategies to generate a multispectral source, supercontinuum generation (SCG), which is the extreme spectral broadening from a pulsed laser source, is suitable for various applications in metrology, imaging, spectroscopy, and optical communications. While initially developed based on optical fibers, tremendous advancements in photonic integrated waveguides have been performed to benefit precise dispersion control, high nonlinearity, and compact size. The fast development of integrated platforms enables new nonlinear processes that go beyond classical SCG, pushing on-chip SCG toward practical applications. This review will begin with the SCG principle in the cubic $\chi^{(3)}$ nonlinear process. Then we analyze recent advances in silicon nitride (SiN) for advanced dispersion, high-nonlinearity III–V platform for efficient SCG and lithium niobate (LN), which exploits both $\chi^{(2)}$ and $\chi^{(3)}$ nonlinearities. Finally, we present a comprehensive discussion of targeted applications, detailing their specific requirements for SCG performance. This analysis provides a forward-looking perspective on the future capabilities of on-chip SCG.

https://doi.org/10.1364/JOSAB.574862

1. INTRODUCTION

Nonlinear optics, ignited by the 1961 observation of second-harmonic generation, has seen continuous growth due to advances in physics, light sources, materials, nanofabrication, and computational power. A nonlinear optical effect, in general, can be exploited to achieve frequency conversion and broadband spectrum generation. Supercontinuum generation (SCG), in which an intense laser pulse undergoes significant spectral broadening while maintaining spatial coherence and brightness, is a very interesting nonlinear optical phenomenon. It has found widespread applications in imaging, optical coherence tomography, spectroscopy, and optical frequency comb technologies. While SCG is rather easy to observe, the control of SCG heavily relies on the linear and nonlinear properties of the optical waveguide.

Photonic waveguides made by CMOS-compatible materials, such as Si, Ge, and SiN, have been initially considered for on-chip SCG due to their mature manufacturing technology. The low-loss waveguide with good optical confinement and highly tailorable dispersion enables efficient frequency conversion to the targeted wavelength range [1–3]. With the recent development of low-loss III–V on insulator platforms, on-chip SCG with even lower pump power requirements has been achieved [4,5]. Furthermore, due to the progress of etching technology, noncentrosymmetric materials, such as lithium niobate (LN), have now been used to take advantage of both $\chi^{(2)}$ and $\chi^{(3)}$ nonlinearities and their cascade effect [6,7]. In parallel, due to the fabrication progress of device geometry, more complex waveguide shapes can be used for SCG such as sign-alternating waveguide [8,9], inverse design waveguide [10], and quasi-phase-matching waveguide [11,12]. These more complex dispersion controls enable relaxed pumping conditions and precise control of spectral broadening.

With the rapid development of on-chip SCG, numerous applications have already been demonstrated, including optical coherence tomography [13], spectroscopy [14–16], calibration of astronomical spectrographs for exoplanet search [17], and high bit-rate telecommunication [18], while other exciting applications, such as imaging and LIDAR, are just on the edge. The gap between the applications' requirements and on-chip SCG performances is rapidly being bridged.

Excellent reviews already address in depth the principles and processes of SCG within silicon photonics [19], as well as comprehensive overviews of on-chip SCG, encompassing its generation, applications, challenges, and future perspectives [20,21]. For a more detailed understanding of the basic physics behind SCG, we recommend consulting these interesting resources. In this review, we aim to cover more recent SCG progress on the integrated platforms. This includes both progress in the material platform and dispersion engineering for a more efficient and tailorable supercontinuum (SC). We also discuss the applications and their specifications from the perspective of on-chip SCG. The review is organized as follows. We start by giving an overview of the theoretical principles of SCG in a $\chi^{(3)}$ photonics waveguide. Then we highlight the advanced dispersion engineering mainly performed in SiN platform, efficient spectral broadening in III–V platform and novel $\chi^{(2)}$ and $\chi^{(3)}$ cascaded effect realized in LN. Finally, we discuss the specifications of SCG used for current applications, aiming to find out the perspective of future SCG on integrated platforms.

2. PRINCIPLE

For most used CMOS-compatible materials, including Si, Ge, and Si_3N_4, only third-order nonlinearity is exhibited, described by the $\chi^{(3)}$, due to their centrosymmetric and isotropic properties. Consequently, an optical pulse in such material can be described by the nonlinear Schrödinger equation (NLSE), assuming a slowly varying pulse envelope. The NLSE, which describes how ultrashort laser pulses travel through materials that both disperse light and exhibit nonlinear optical properties, is usually expressed in a reference frame moving at group velocity of the pulse. The NLSE is expressed as [22]

$$\frac{\partial A}{\partial z} = -\frac{\alpha}{2}A + i\sum_{k\geq 2}\beta_k \frac{i^k}{k!}\frac{\partial^k A}{\partial t^k} + i\gamma\left(1 + i\tau_{\text{shock}}\frac{\partial}{\partial t}\right)|A|^2 A, \tag{1}$$

where A is the temporal envelope of the pulse. α is the propagation loss. Key to this process are fundamental phenomena like group velocity dispersion (GVD, β_2), higher-order dispersion (β_k, $k > 2$), and self-phase modulation (SPM) introduced by the nonlinear coefficient γ. γ is frequency-dependent, which is modeled to the first order by τ_{shock}. Here, we ignore the Raman effect because integrated photonic waveguides are usually short, leading to negligible Raman phenomena.

The most common configuration for SCG is taking advantage of the optical soliton, requiring the pump locates at the anomalous GVD (AD, $\beta_2 < 0$) regime of the waveguide. The optical soliton will go through the soliton self-compression stage and, if given the proper higher-order dispersion, phase match with quasi-continuous waves, noted as dispersive waves (DWs). The frequency of the dispersive wave can be predicted by the phase-matching condition [22], as follows:

$$\beta_{\text{int}}(\omega) \equiv \beta(\omega) - \beta(\omega_p) - (\omega - \omega_0)v_g^{-1} = 0. \tag{2}$$

The soliton dynamics and DW generation lead to wide spectral broadening. Figure 1 illustrates a typical SCG process in a buried photonic SiN-on-insulator waveguide, the cross section of which is shown in Fig. 1(a). Based on this cross section, the GVD is simulated and depicted by the purple curve in Fig. 1(c). The phase-matching condition for DW generation is represented by the black curve in the same figure. A pump laser at 1380 nm, located in the anomalous dispersion (AD) region, is used to initiate the SCG. Numerical simulations using the NLSE [Fig. 1(b)] show a significant spectral broadening from 600 to 1900 nm. The two distinct DWs generated in this process are identified by the green lines, which correspond precisely to the graphical solutions of the phase-matching condition.

3. ADVANCED DISPERSION ENGINEERING IN SILICON NITRIDE

Dispersion engineering is crucial for SCG, as it directly controls the pulse evolution within a waveguide, while material nonlinearity typically remains relatively constant during propagation. The well-established fabrication processes for Si and SiN enable precise control over waveguide geometry. This precision facilitates the exploration of advanced dispersion engineering techniques, which involve the variation of the waveguide dispersion along the propagation direction. Consequently, this allows for more precise control over both pulse evolution and spectral broadening. Especially when pumping in the near-infrared range, SiN is clearly the preferred material due to its absence

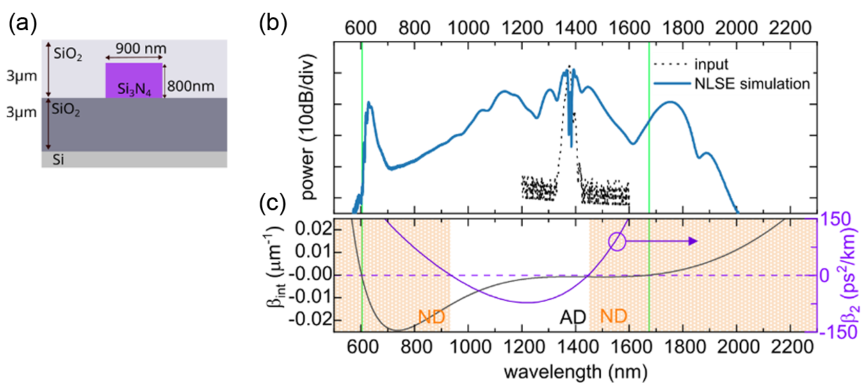

Fig. 1. Typical SCG process in a buried SiN-on-insulator photonic waveguide (a) The cross-section (800 nm height and 900 nm width) of the waveguide on which the simulation is based. (b) Spectral broadening in blue from NLSE simulation. Pump condition: 190 fs squared hyperbolic secant pulse. Peak power $P_0 = 290$ W at 1380 nm wavelength. The initial pump pulse spectrum is shown in the dotted black curve. Propagation distance 2.3 cm. (c) GVD β_2 curve in purple shows a hyperbolic shape, indicating a non-negligible β_3 and β_4 contribution. β_{int} curve is shown in the black curve, indicating two phase-matching wavelengths (in vertical green lines).

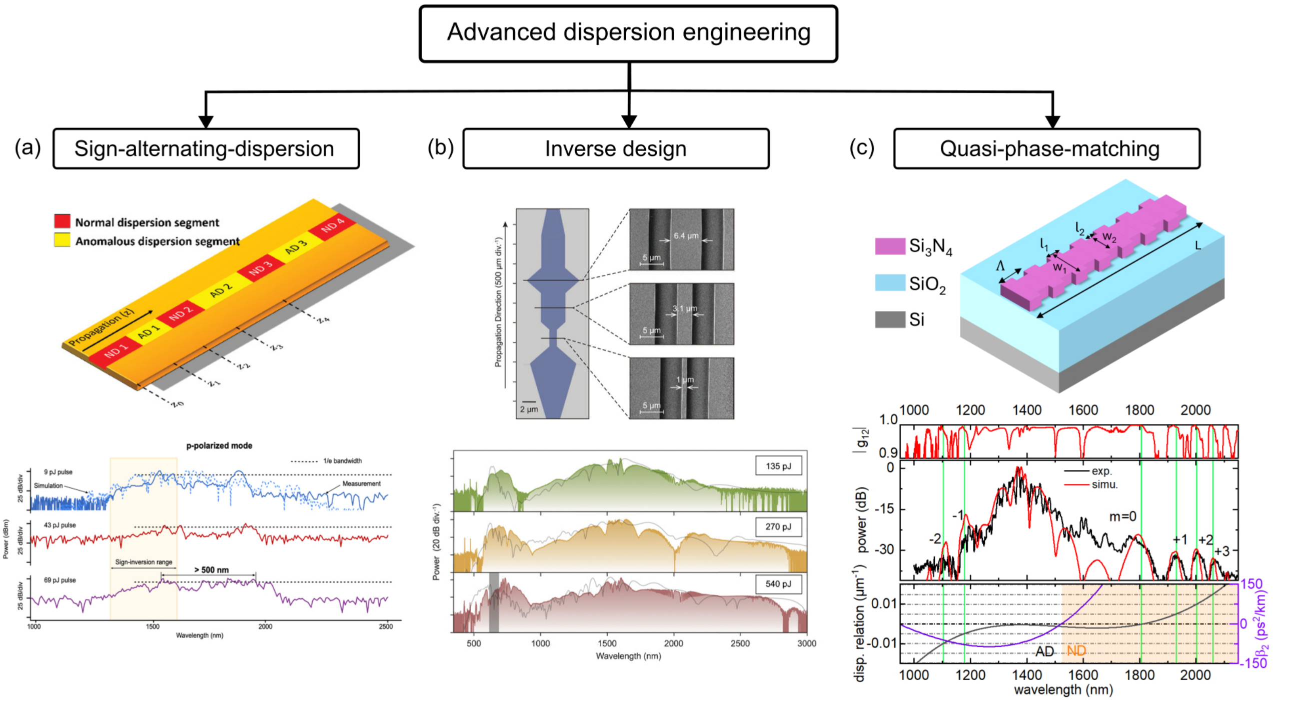

Fig. 2. Three advanced dispersion engineering techniques of which the waveguide dispersion changes with pulse propagation. (a) Sign-alternating-dispersion waveguide for efficient spectral broadening [8,9] at −4 dB range. (b) Inverse design waveguide for flat and coherent spectra [10]. (c) Periodic varying waveguide for a highly tailorable quasi-phase-matching SCG [12].

of two-photon absorption, ensuring freedom from nonlinear loss within the waveguide. Here, we identify three methods: sign-alternating waveguide, inverse design waveguide, and quasi-phase-matching (QPM) waveguide.

The sign-alternating waveguide was first designed and demonstrated in [9] as presented in Fig. 2(a). The waveguide is composed of different segments to obtain an ultra-efficient spectral broadening by alternating the normal dispersion (ND) and anomalous dispersion (AD) [8]. Specifically, the loss of peak power caused by the temporal broadening in ND segments is recovered by the pulse compression in the following AD segments. Conversely, the spectral broadening that may cease in AD segments due to soliton formation is rebroadened again by the spectral generation in ND sections. This continuous cycle allows spectral generation to persist over a longer waveguide distance, leading to significant improvement in the SCG bandwidth. A bandwidth of 950 nm was achieved with only 34 pJ of input energy, and a bandwidth of 375 nm was achieved with only 9 pJ. This is a performance metric at a high spectral power level (−4 dB), which is crucial for applications such as optical coherence tomography (OCT).

Inverse design has also been applied to SCG by unlocking the geometric degrees of freedom in waveguide design that were previously under-evaluated due to the complexity of the underlying nonlinear physics. In a recent demonstration, spectrum flatness and temporal coherence were used as a figure of merit (FOM) during the optimization process [10]. The authors use a particle swarm optimization (PSO) algorithm, which is insensitive to initial values and can avoid numerous local minima. Moreover, its compatibility with parallel computing drastically accelerates the computationally intensive design process. Through simulations, the authors found that inverse design can effectively manage pulse temporal overlap. This promotes spectral broadening through cross-phase modulation and wave-trapping phenomena that are difficult to control with conventional forward design approaches. Experimentally, the authors have successfully demonstrated a flatter SC in the mid-infrared range [10] [Fig. 2(b)].

Meanwhile, the implementation of the quasi-phase-matching (QPM) approach, achieved through the modulation of dispersion along the waveguide propagation, facilitates the generation of multiple DWs [11,12]. Different from the established QPM scheme in $\chi^{(2)}$ material, which typically involves periodic changes in nonlinear polarizability for efficient harmonic generation, the QPM employed for DW generation introduces a significantly longer periodicity. This extended periodicity serves to compensate for the phase mismatch between solitons and DWs. The wavelength of QPM-DWs can be predicted from the QPM condition [23], knowing the modulation period Λ:

$$\beta_{\mathrm{int}}(\omega) = \frac{2\pi}{\Lambda}. \tag{3}$$

As shown in Fig. 2(c), SCG with multiple QPM-DWs is achieved in a buried SiN waveguide [12]. The waveguide is 800 nm thick, and its width varies between 700 and 1100 nm with a period of 2.5 mm. This waveguide is pumped with a laser at 1380 nm, which is in the anomalous dispersion region. The varying width of the design facilitates the generation of multiple QPM-DWs, which are clearly visible in the resulting SCG spectrum.

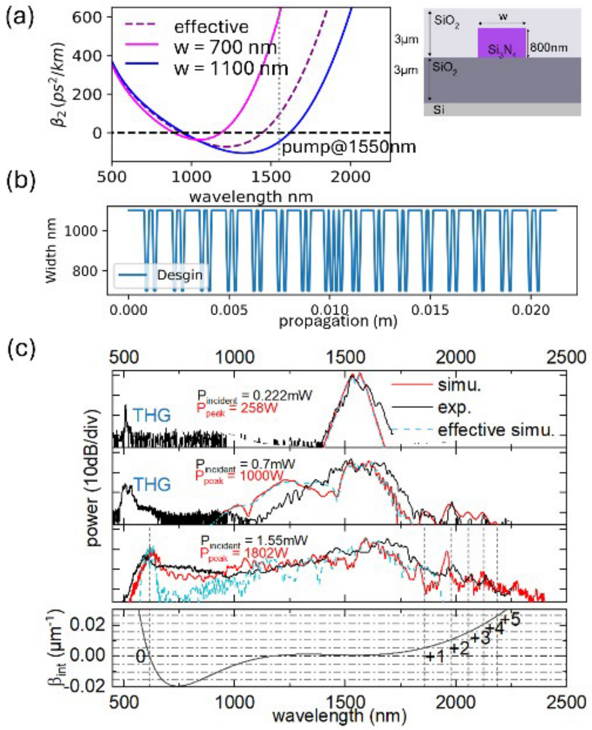

Fig. 3. QPM-DW generation under a pumping in normal dispersion region. (a) Dispersion curves and cross sections for the waveguide geometries. (b) Longitudinal evolutions of waveguide width. The period Λ is centered at 1.3 mm. (c) Experiment and simulation results under a 190 fs TE pump at a wavelength of 1550 nm. Measurements and simulations based on the QPM waveguide are shown in black and red, respectively, while the simulation based on the effective dispersion model is shown in light blue. Please note that the THG signal and IR signal should not be compared in intensity since they are detected by two sensors with different noise floors. Different orders of QPM-DWs predicted from a graphic solution of the QPM condition.

It is worth noting that DWs can be generated not only when the pump is in the anomalous dispersion region, but also when the pump is in the normal dispersion region [24,25]. This phenomenon can be further enhanced by QPM, leading to the generation of QPM-DWs even under normal dispersion conditions.

An illustrative example of this QPM effect is demonstrated in a SiN waveguide designed for a 1550 nm TE mode pump. As detailed in Fig. 3, the 800 nm thick, 2.1 cm long spiral waveguide [Fig. 3(a)] incorporates a slightly chirped periodic variation in width, transitioning from 1100 to 700 nm along its propagation axis [Fig. 3(b)]. The effective dispersion, representing the average dispersion experienced by the propagating pulse, is depicted by the dashed purple curve in Fig. 3(a). In the setup, a 190 fs pump laser with a 1 MHz repetition rate, centered at 1550 nm, is situated within the normal dispersion region.

Figure 3(c) presents the spectral broadening observed in both experimental and simulated results. As the pump power increases, the spectral broadening progresses from initial contributions by SPM and third-harmonic generation (THG) to the emergence of multiple QPM-DW peaks. In particular, QPM-DWs of orders ranging from 0 to +5 are clearly observed. The excellent agreement between the experimentally observed DW locations and the QPM condition, derived from Eq. (3) also shown in Fig. 3(c), confirms that these dispersive waves are precisely determined by the QPM terms.

The QPM method provides an additional way to trigger the energy transition to user-defined spectral ranges, which can be served as a promising multicolor source.

Recent advances in advanced dispersion engineering are paving the way for the future of SCG. These new techniques allow for more precise control over both pulse evolution and the extent of spectral broadening. While these explorations have shown great promise in the SiN platform, they can be further extended to other material platforms to leverage their higher nonlinearity or different transparency windows for a wider range of applications.

4. EFFICIENT SPECTRAL BROADENING IN III–V PLATFORMS

Although the nonlinearity of an integrated platform, such as SiN, is already one to two orders of magnitude higher than that of optical fibers, a the typical pump peak power of hundreds to thousands of watts is still required, bringing a significant challenge to the realization of a fully integrated SC system [20]. The main reason for this is that the propagation distance available for the integrated platform is limited by the propagation loss or the chip footprint. Therefore, the development of novel integrated platforms with low loss and even higher nonlinearity is an urgent need. In this context, there has been an increasing interest in III–V platforms in recent years.

The high nonlinear index and strong mode confinement due to a high refractive index contrast together result in a nonlinearity that is one or two orders of magnitude higher than the Si_3N_4 platform [26]. Especially, AlGaAs has almost the highest $\chi^{(3)}$ nonlinearity among all the integrated platforms, drawing much attention for nonlinear applications. The first octave spanning SC was realized in [27] on an AlGaAs-on-insulator platform in telecom wavelength. This was achieved by using a 100 fs pump with remarkably low pulse energy of only 3.6 pJ, corresponding to an estimated peak power of 32 W. Recent demonstrations also showed the potential to combine both $\chi^{(2)}$ and $\chi^{(3)}$ effects together with high fabrication quality with dB/cm level loss [4,28]. A demonstration of 1.2 octave SCG has been achieved on 400 nm height and 800 nm wide AlGaAs-on-insulator waveguide [see Fig. 4(a)] [4]. This was accomplished using a 100 fs pump centered at 1.95 μm with pulse energy of 12 pJ, which corresponds to an estimated peak power of 100 W. This SCG was made possible by leveraging both $\chi^{(2)}$ and $\chi^{(3)}$ nonlinearities within the waveguide.

Recent demonstration of GaAsSb/InP waveguide showed a wide SCG in mid-infrared pumped at a wavelength of 5 μm [Fig. 4(b)] [5]. The 2 μm thick GaAsSb core has both upper and lower cladding of InP, which are 2 and 4.5 μm thick, respectively. The waveguide is 20 mm long and 5.5 μm wide. Important spectral broadening has been observed with only a peak power of 6–45 W with 200 fs pump. The results pave the way for developing an electrically pumped, chip-scale mid-infrared supercontinuum source by monolithically integrating a quantum cascade laser (QCL) pump with passive waveguides on an InP platform.

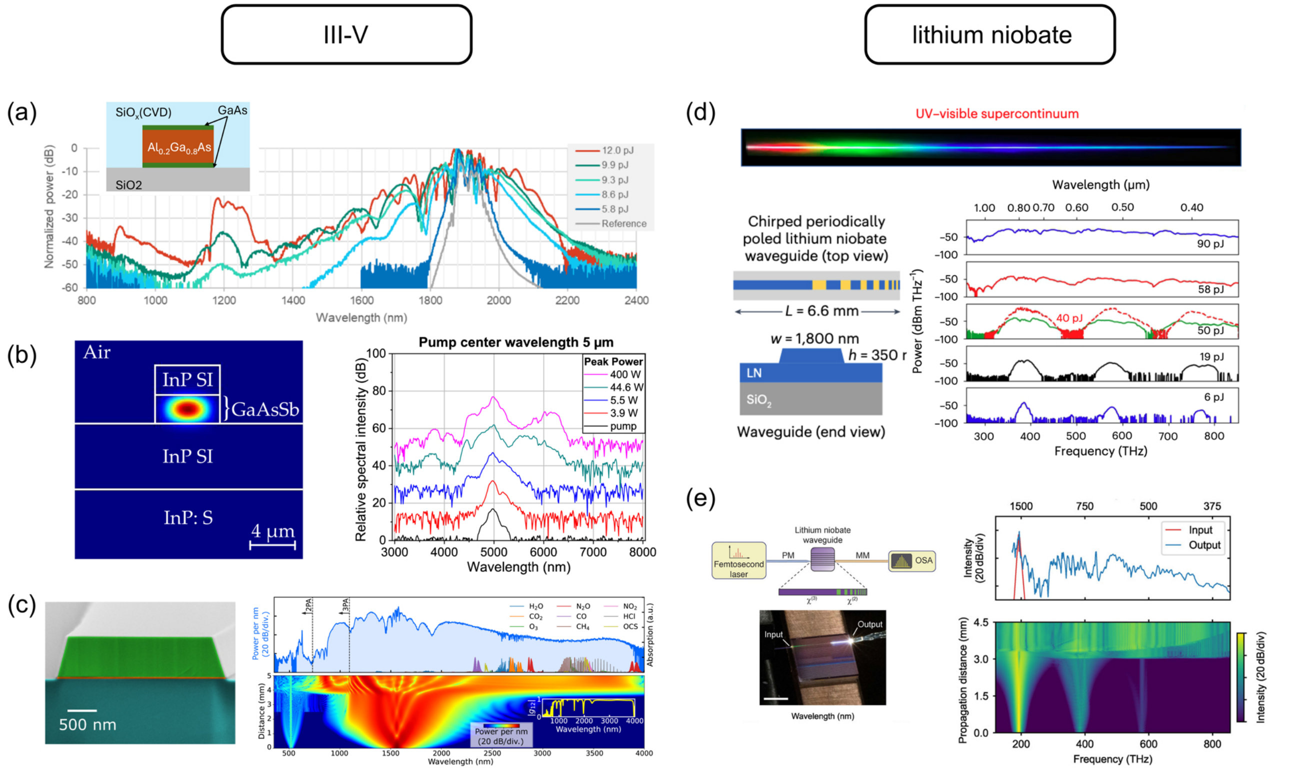

Fig. 4. Recent SCG achievements on platforms III–V and lithium niobate. Demonstrations in AlGaAs-on-insulator (© 2024 IEEE; reprinted with permission from [4]) pumped at 1.9 µm nm wavelength, in GaAsSb/InP [5] pumped at 5 µm, and in GaN-on-sapphire [26] pumped at 1.5 µm are shown in (a), (b), and (c), respectively. (d) SCG demonstration in chirped periodic poling lithium niobate (PPLN) pumped at wavelength at 0.8 µm. The spectral broadening is a result from both $\chi^{(2)}$ and $\chi^{(3)}$ nonlinearities. (e) SCG demonstration in a fully etched chirped PPLN pumped at wavelength at 1.56 µm.

Broadband supercontinuum generation has also been realized in 725 nm gallium nitride on sapphire (GaN-sapphire) waveguides [Fig. 4(c)] [26]. GaN, which benefits from a large band gap, has no two-photon or three-photon absorption for a pump located at the telecom C-band (around 1550 nm). The demonstration showed the potential of GaN for creating mid-infrared up to 4 µm light from near-infrared lasers measured from a 2-segment 5-mm waveguide whose widths change from 4 µm to 2 µm. This was achieved with a 55 fs pump pulse with 527 pJ of on-chip energy, corresponding to an estimated peak power of 8.3 kW. This work highlights the strong potential of GaN for creating on-chip MIR light sources.

While III–V platforms offer superior nonlinearity for SCG, their application is currently limited by practical challenges. These include waveguide lengths, which are typically limited to a few millimeters, and the difficulty of achieving high coupling efficiency due to tight mode confinement. Future research should focus on improving fabrication quality to enable longer waveguides and developing techniques for higher coupling efficiency. Successfully addressing these issues will pave the way for a significant reduction in the required pumping power for on-chip SCG.

5. PROGRESS IN THIN-FILM LITHIUM NIOBATE WITH BOTH $\chi^{(2)}$ and $\chi^{(3)}$ EFFECTS

Although SCG mainly depends on the $\chi^{(3)}$ nonlinearity, the possibility of having $\chi^{(2)}$ for second harmonic generation in the same platform is of great interest, especially for $f-2f$ interferometry. For the centrosymmetric material like SiN, $\chi^{(2)}$ effect can be artificially induced by the photogalvanic effect [29,30], but this optical poling can be erased by the high-energy photons present in the SCG process [31]. A more straightforward approach is to use materials with an intrinsic $\chi^{(2)}$ nonlinearity, such as lithium niobate (LN) and III–V materials such as AlGaAs. Furthermore, $\chi^{(2)}$ can enable additional nonlinear processes that enhance SCG. It allows for spectral broadening into longer wavelengths by difference frequency generation (DFG) and into shorter wavelengths through second-harmonic generation (SHG). It can also produce a strong effective SPM under phase-mismatched conditions [7], which further contributes to the spectral broadening.

In particular, the wide transparency window of LN is a significant advantage as it enables spectral broadening that spans the UV and visible ranges. This capability is crucial for applications in quantum technologies [32,33], microscopy [34], and astronomy [17,35]. An important approach to leveraging LN's properties is the use of periodically poled lithium niobate (PPLN) waveguides. These waveguides take advantage of both the material's strong intrinsic $\chi^{(2)}$ and $\chi^{(3)}$ nonlinearities,

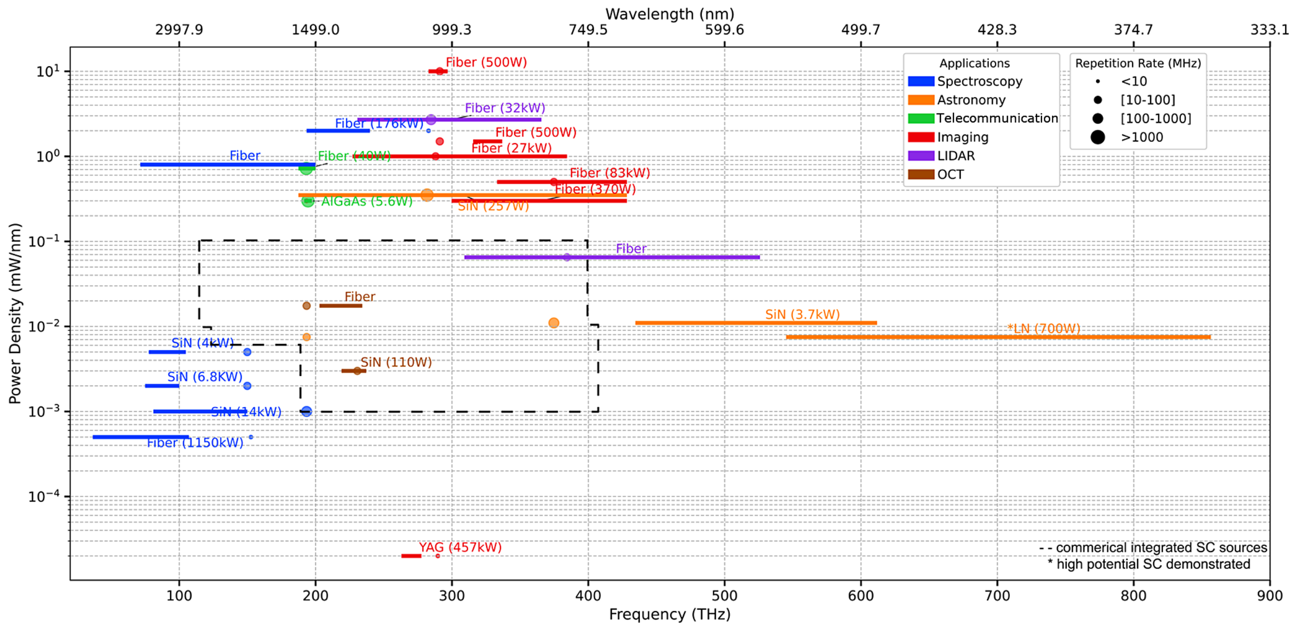

Fig. 5. Overview of the experimental generation and use of SC across application domains. The position of the circular markers indicates the pump wavelength, and the bar indicates the range used in the application. The color indicates the application, and the marker size shows the repetition rate of the pump laser. The platform and peak power of the pump laser source are indicated for each range. The properties of commercially available SC sources based on integrated waveguides are noted within the range indicated by the dashed line.

along with their cascading effects. As shown in [6] [Fig. 4(d)], one such design uses a two-stage process. The 350 nm shallow-etched waveguide guide is equipped with a first-stage unpoled waveguide to broaden the initial spectrum around the pump wavelength at 1550 nm, followed by the second stage where the poling period varies from 12.5 to 2.5 μm. The chirped poling enables the broadband quasi-phase matching and the wideband harmonic generation. The resulting harmonics overlap to create continuous spectral broadening, efficiently extending the light into the visible and UV regions.

A similar strategy has also been applied to a fully etched thick PPLN (800 nm) [Fig. 4(e)]. This improved geometry provides better control over the dispersion characteristics, enabling a multi-octave spectral broadening from the pump wavelength 1560 nm to the UV [36].

6. APPLICATIONS

Following recent advances in on-chip SCG, both in materials progress and in different advanced dispersion engineering techniques, we present an overview of the SC applications, focusing mostly on recent demonstrations with SC sources realized in integrated photonic platforms. To show the future potential applications for on-chip SCG, we also include in application demonstrations in the fiber and bulk material. Indeed, most of the SCG demonstrate on integrated platforms focus on spectral broadening and temporal coherence, whereas from an application point of view, other specifications are equally important, including repetition rate and power density. Here, we focus on the main showcase SC applications: spectroscopy, optical coherence tomography (OCT), astronomy, telecommunication, imaging, and LIDAR. We provide an overview of the pumping wavelength, the spectral broadening range, the repetition rate, and the power density in the targeted spectral range.

Figure 5 reports different showcases of applications using SCG. The details of each reported demonstration are presented in Table 1. The figure is organized by showing spectral range (in frequency and wavelength) with power density (in mW/nm) that is used for the application demonstration. Different applications are plotted in different colors. The pumping conditions, such as pump wavelength and pulse repetition rate, are indicated by the circle dots and their diameters, respectively. The platforms for SCG and the initial peak powers are also noted. The spectral broadenings are drawn in lines. We only focus on the specific spectral broadening that is useful, which is typically a much narrower range than the total SC spectrum. The power densities are estimated by average power over the corresponding spectral broadening, as most of the demonstrations did not indicate the exact power densities. While this estimation may not be accurate because the SC is not necessarily flat, it gives a direct insight into the power requirement per unit bandwidth for different applications.

The mid-infrared (MIR) spectral range is particularly well-suited for spectroscopy of chemical and biological substances due to the characteristic molecular absorption fingerprints within this region. In the telecommunications band, specifically around 1550 nm, flat supercontinua offer direct application as hundreds of parallel channels for high-capacity data transmission. Moving into the near-infrared (NIR) range, its utility extends to applications such as optical coherence tomography (OCT), various imaging techniques, LIDAR systems, and astronomical observations. Furthermore, the visible and ultraviolet (UV) ranges present significant unexplored potential

Table 1. SCG Applications across Various Fields

				Pump Conditions					Supercontinuum Characteristics		
Application	**Ref.**	**Year**	**Platform**	ω_p(μm)	$\tau(fs)$	E	P_p	R(MHz)	ω_s(μm)	ω_e(μm)	D_P(mW/nm)
Spectroscopy	[37]	2014	Fiber	1.06	5 ps	1 μJ	176 kW	0.7	1.25	1.55	2
	[38]	2019	Fiber	1.55	\	\	\	2.5	1.50	4.20	0.05–0.8
	[39]	2024	Fiber	1.965	765	1 μJ	1150 kW	1	2.80	8.20	0.0005
	[14]	2019	SiN	2.00	78	750 pJ	6.8 kW	19	3.00	4.00	0.002
	[16]	2020	SiN	2.00	90	420 pJ	4 kW	20	2.86	3.86	0.005
	[15]	2020	SiN	1.55	70	1 nJ	14 kW	250	2.00	3.70	0.001
OCT	[13]	2021	SiN	1.30	200	25 pJ	110 W	80	1.264	1.369	0.003
	[40]	2021	Fiber	1.55	125	\	\	90	1.28	1.478	0.0175
Astronomy	[6]	2024	LN	1.55	130	90 pJ	700 W	100	0.35	0.55	0.0075
	[17]	2019	SiN	1.064	70	18 pJ	257 W	30 GHz	0.70	1.60	0.35
	[35]	2018	Fiber	0.80	27	100 pJ	3.7 kW	1 GHz	0.49	0.69	0.011
Imaging	[41]	2016	Fiber	1.55	1 ns	370 nJ	370 W	0.25	0.70	1.00	0.3
	[42]	2023	Fiber	0.80	120	10 nJ	83 kW	80	0.70	0.90	0.5
	[43]	2016	Fiber	1.041	220	6 nJ	27 kW	80.2	0.78	1.32	1
	[44]	2024	YAG Crystal	1.035	350	160 nJ	457 kW	2.5	1.08	1.14	0.02
	[45]	2020	Fiber	1.03	350	175 pJ	500 W	20	0.89	0.95	1.5
									1.01	1.06	10
Tele-communication	[18]	2018	AlGaAs-OI	1.542	1.5 ps	8.5 pJ	5.6 W	10 GHz	1.52	1.564	0.3
	[46]	2015	Fiber	1.55	\	\	40 W	6.25 GHz	1.50	1.60	0.72
LIDAR	[47]	2023	Fiber	0.78	\	\	\	100	0.57	0.97	0.065
	[48]	2025	Fiber	1.053	100	3.2 nJ	32 kW	1 GHz	0.82	1.30	2.7

for advancements in astronomy [17], microscopy [34], and emerging quantum applications [32].

The required power density of a SC source varies significantly across different applications. Spectroscopy and OCT, for instance, typically operate efficiently at relatively low power densities (10^{-2} to 10^{-3} mW/nm). This characteristic has facilitated their successful demonstration on integrated photonics platforms. In contrast, imaging applications often require higher power densities (usually above 10^{-1} mW/nm), particularly when they involve excitation processes or nonlinear frequency conversion. Consequently, these applications have predominantly relied on fiber- or bulk material-based SCG, as integrated platforms have historically faced challenges in achieving the required power levels. Similarly, applications, such as LIDAR and telecommunications, which involve long-distance propagation, also benefit considerably from higher power densities to ensure signal integrity and reach.

Certain applications are critically dependent on high-repetition-rate SCG. For instance, ultradense wavelength-division multiplexing (WDM) in telecommunications needs a frequency spacing of 12.5 GHz. To support this, a coherent SCG source must similarly operate at this repetition rate. Furthermore, astronomers require frequency combs with line spacing typically ranging from 10 to 30 GHz [17]. These specialized sources are crucial for various astronomical applications.

It is worth noting that commercial SC sources based on integrated waveguides are already available [49,50]. While integrated waveguides provide strong nonlinear effects, these systems still rely heavily on fiber components for functions such as pump lasers, amplification, and dispersion management. The properties of such SC sources are shown in Fig. 5 (dashed region), with repetition rates ranging from 100 MHz to 30 GHz. These specifications already meet the requirements for OCT. However, many other potential applications remain out of reach.

For future applications of SCG with an integrated platform, several opportunities are emerging. First, recent demonstrations of SCG extending into the visible and UV spectral ranges [6,36] (see $*$ case in Fig. 5) offer promising opportunities for unexplored applications in both astronomy and quantum technologies. Second, while on-chip SCG has achieved spectral extension to wavelengths at or beyond 10 μm [3,5,51], practical spectroscopic demonstrations leveraging these broad MIR spectral components have not been achieved. Future efforts can be made on showcasing these power-efficient MIR sources enabled by integrated photonics through concrete spectroscopic applications. Third, integrated platforms are also suitable for delivering high repetition rate and high-power density SCG, although most demonstrations to date remain below the gigahertz range. Specifically, waveguides fabricated from materials like silicon nitride or lithium niobate, known for their high nonlinearity and broad transparent windows, can generate SC with minimal pulse energy. This capability facilitates repetition rates exceeding 10 GHz while maintaining average power levels within typical amplification limits (e.g., a few watts). Such advances will unlock new possibilities in diverse fields, including astronomy, quantum technologies, telecommunications, and high-speed sensing.

7. CONCLUSION

Significant efforts have been dedicated to developing on-chip supercontinuum generation, focusing on both dispersion engineering and novel material platforms to meet various application requirements. Dispersion engineering has been widely used in shaping the spectral broadening of SC. Recently, the need for more precise control of SCG has driven the advanced dispersion engineering. We have identified three methodologies in this domain: sign-alternating waveguides, inverse-designed waveguides, and quasi-phase-matching waveguides. These approaches collectively represent promising avenues for optimizing SCG performance and tailoring spectral output for diverse applications.

While silicon-compatible materials have long dominated on-chip supercontinuum generation due to mature manufacturing processes, there is a growing interest in III–V platforms driven by their exceptionally high nonlinearity. Significant progress has led to the realization of low-loss passive devices made from III–V materials. Recent demonstrations of SCG on these platforms highlight their potential for reduced pump power requirements and the future development of fully integrated SCG systems.

At the same time, breakthroughs in the fabrication of thin-film lithium niobate (LN) have led to the ability to use both second- and third-order nonlinearities for SCG. These rich nonlinear interactions offer a promising pathway to highly efficient SCG, particularly extending into the visible and ultraviolet (UV) spectral regions.

Despite significant advances in on-chip SCG, important challenges remain before fully integrated SC sources can address the full spectrum of applications.

A central limitation is power density. While low-power-demand applications, such as spectroscopy and optical coherence tomography (OCT), have clearly demonstrated the promise of on-chip SCG, extending its use to more demanding applications like imaging will require higher power levels. Future progress is expected from the use of high-repetition-rate pump sources, such as electro-optic combs [17,52], which can significantly boost the power density. Such developments could unlock the potential of on-chip SCG for imaging, astronomy, and telecommunications.

Another key challenge lies in the reliance on external pump lasers, which limits long-term stability, scalability, and portability. A fully integrated SCG platform that incorporates an on-chip pump laser would represent a major step forward. Although most current integrated mode-locked lasers—whether based on monolithic III–V integration [53] or heterogeneous silicon nitride platforms [54]—still deliver only a few picojoules of pulse energy, recent progress in doped waveguides points toward a solution [55,56]. Integrated *Q*-switched and mode-locked lasers have already achieved pulse energies comparable to those of fiber-based systems [57,58], suggesting that fully integrated SC sources with sufficient power are within reach. Achieving this vision would open the door to compact, robust, and widely deployable SC systems with impact across diverse scientific and technological domains.

Funding. European Research Council (ERC-CRYPTONIT, 101097804); Agence Nationale de la Recherche (CARBONIC, ANR-21-CE09-0023).

Disclosures. The authors declare no conflicts of interest.

Data availability. The data re-analyzed in this review are available within the cited literature. The new data generated and analyzed during this study are available in Zenodo at [59].

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